

Auditing the RTE Compliance Gap: Network-Based Geospatial Analysis of Walking Access to Government Schools in Bengaluru’s Formal and Informal Settlements

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Abstract—The Right of Children to Free and Compulsory Education (RTE) Act, 2009, and its rules require a government primary school within 1 km walking distance of every child’s habitation. Compliance is usually audited with outdated census frames and straight-line (“crow-flies”) buffers, which are blind both to informal settlements and to the real street network. This paper presents an open, reproducible geospatial pipeline that audits RTE walking-distance compliance in Bengaluru. Sentinel-2 imagery and a Random Forest classifier delineate formal and informal built-up morphology; OpenStreetMap (OSM) supplies a 186,595-node pedestrian graph and 1,464 school features; and a single multi-source Dijkstra search, which partitions the street network into government-school catchments (a network Voronoi diagram), computes the exact network distance from 1,000 classified sample locations and all 509 Karnataka Slum Development Board (KSDB) slum polygons to the nearest government school. WorldPop 2020 population estimates weight every result. Using a curated set of 111 government schools, only 17.3% of sampled locations and 20.1% of the 226,753 slum residents lie within a 1 km walk. Planned layouts are on average 220 m farther than informal areas (2.32 vs. 2.10 km), but the difference is negligible in effect size (Cliff’s $\delta = 0.06$) and sensitive to how government schools are identified. Crow-flies buffers overstate compliance: 48.2% of locations that appear compliant by straight line are non-compliant by network distance. Non-notified slums are significantly farther from government schools than notified slums (population-weighted 2.40 vs. 1.71 km; $p < 10^{-5}$), and slum demand is highly concentrated: the ten most-loaded schools are the nearest government school for 71.5% of slum residents (Gini = 0.85). Meanwhile, 71–76% of locations lie within 1 km of a non-government school. The results show that the compliance gap is city-wide, that it is underestimated by Euclidean audits, and that it falls hardest on settlements that are administratively invisible. Because OSM under-records government schools, all compliance figures are best read as lower bounds on access.

Index Terms—Right to Education, spatial accessibility, informal settlements, OpenStreetMap, OSMnx, network analysis,

Dijkstra, Sentinel-2, Random Forest, WorldPop, Bengaluru.

I. INTRODUCTION

A. Problem Domain

Bengaluru is one of the fastest-growing metropolitan regions in India. The 2011 Census recorded 8.44 million residents within the Bruhat Bengaluru Mahanagara Palike (BBMP) limits and 8.52 million in the urban agglomeration [1], and growth since then has been concentrated at the periphery, much of it unplanned. Rapid growth of this kind produces informal settlements whose boundaries, populations, and service needs are not reflected in the planning record for years [2]. In Bengaluru, the KSDB maintains a slum register, but 317 of the 509 slum polygons in its published layer are *non-notified*, i.e., recognised spatially but without the legal status that triggers service provision.

The RTE Act, 2009 [3], together with the Central RTE Rules, 2010 [4], (Rule 6) defines the “neighbourhood” for a primary school (Classes I–V) as an area within a walking distance of 1 km, and for an upper-primary school (Classes VI–VIII) as within 3 km. Compliance is normally assessed from habitation lists and straight-line radii. Two failure modes follow. First, a settlement that is absent from the planning frame cannot be audited at all. Second, a straight-line buffer ignores the street network: a school 600 m away across a railway line, an arterial road without crossings, or a walled campus may be a 2 km walk. We call the second failure mode the *crow-flies fallacy*.

B. Motivation

Open Earth-observation and volunteered geographic data now make it possible to audit access continuously and at city scale. Sentinel-2 provides free 10 m multispectral imagery every few days [5]; Google Earth Engine (GEE) makes classification over a metropolitan area tractable [6]; OSM provides a routable street network [7]; and WorldPop provides gridded population estimates at 100 m [8]. Combining these sources gives planners a way to measure what the RTE mandate actually guarantees—a walk, not a radius.

C. Contributions

This paper makes five contributions:

- 1) An open, end-to-end pipeline (Map → Detect → Measure → Weigh) that combines Sentinel-2 classification, OSM network routing, KSDB slum polygons and WorldPop population to audit RTE walking-distance compliance (Section IV).
- 2) An application of network Voronoi partitioning [23], [24] to RTE catchments: a single multi-source shortest-path search returns, for every street node, both the walking distance to the nearest government school and the identity of that school, enabling catchment maps and population-weighted catchment-load analysis (Algorithm 1).
- 3) A quantitative test of the crow-flies fallacy on real data, showing that Euclidean buffers roughly double the apparent compliance rate (Section V-D).
- 4) Population-weighted accessibility for all 509 KSDB slums, disaggregated by administrative zone and notification status (Section V-E).
- 5) A transparent treatment of data-quality threats—school-type misclassification in OSM and weak agreement between the satellite classifier and KSDB ground truth—including a sensitivity analysis on the definition of a “government school” (Section VII).

D. Paper Organisation

Section II reviews related work. Section III describes the study area and data. Section IV presents the methodology. Section V reports results, Section VI discusses them, Section VII addresses threats to validity, and Section VIII concludes.

II. LITERATURE SURVEY

A. Remote Sensing of Informal Settlements

Kuffer *et al.* [9] reviewed fifteen years of slum mapping from space and identified image texture, morphology, and object-based image analysis as the most consistent discriminators of informal from formal built-up areas, with very-high-resolution (VHR) imagery dominating the literature. Kohli *et al.* [10] formalised these discriminators in a generic slum ontology (environs, settlement, and object levels) intended to make image-based classification transferable. Mahabir *et al.* [11] surveyed high- and VHR approaches and highlighted the scarcity of reliable ground truth and the cost of VHR data as persistent obstacles. Duque *et al.* [12] showed that

land-cover and Grey-Level Co-occurrence Matrix (GLCM) texture metrics [13] derived from imagery explain a substantial share of intra-urban variation in poverty. Wurm *et al.* [14] applied transfer learning with fully convolutional networks to QuickBird and Sentinel-2 imagery of Mumbai and found that slum segmentation from 10 m Sentinel-2 data is possible but markedly less accurate than from VHR data.

B. Classifiers and Spectral Indices

Random Forests [15] are the most widely used classifiers in operational remote sensing because they handle correlated features, need little tuning, and resist overfitting [16]. Spectral indices are standard input features: the Normalised Difference Vegetation Index (NDVI) [17] separates vegetation from built surfaces, and the Normalised Difference Built-up Index (NDBI) [18] highlights impervious cover using the short-wave infrared band.

C. Spatial Accessibility Measurement

Talen and Anselin [19] showed that conclusions about the spatial equity of public facilities depend strongly on the accessibility measure chosen. Luo and Wang [20] introduced the two-step floating catchment area (2SFCA) method, which accounts for both supply capacity and demand within a travel-distance catchment. Radke and Mu [21] proposed the spatial-decomposition approach that 2SFCA builds upon. Network distance is preferred over Euclidean distance wherever a routable network exists; OSMnx [7] made it practical to download, simplify and analyse OSM street networks for whole cities, and NetworkX [22] provides the shortest-path algorithms, including Dijkstra’s [25]. Assigning every point of a network to its nearest facility produces a network Voronoi diagram, whose theory and computation are reviewed by Erwig [23] and Okabe *et al.* [24].

D. Open Data Quality

OSM completeness varies by place and feature type. Haklay [26] found OSM road positional accuracy in England to be comparable to the national mapping agency in well-mapped areas, but uneven elsewhere. Barrington-Leigh and Millard-Ball [27] estimated that the global OSM road network was over 80% complete by 2016, with large Indian cities among the better-mapped regions. Points of interest such as schools, and especially their attributes (ownership, level), are considerably less complete.

E. Private Schooling in Urban India

A large literature documents the growth of low-fee private schooling in India. Kingdon [28] reviewed its scale and drivers, and Muralidharan and Sundararaman [29] provided experimental evidence on school choice. This context matters for interpreting distance to *government* schools: where private schools are dense, distance to a government school measures the reach of the free public option rather than access to any school at all.

TABLE I
COMPARISON OF RELATED WORK

Study	Focus	Data	Method	Gap relative to this work
Kuffer [9]	Review of slum mapping	VHR, medium-resolution imagery	Texture, OBIA, ML	Detection only; no service access
Kohli [10]	Slum ontology	VHR imagery	Rule-based ontology	No routing or population weighting
Duque [12]	Intra-urban poverty	VHR land cover	Texture metrics, regression	Poverty proxy, not service access
Wurm [14]	Slum segmentation, Mumbai	QuickBird, Sentinel-2	FCN transfer learning	No link to public services
Talen [19]	Equity of playgrounds	Facility points, census	Several accessibility measures	Census frame misses informal areas
Luo [20]	Health-care access	Facilities, census tracts	2SFCA	Requires reliable capacity data
Boeing [7]	Street-network tooling	OSM	Graph analytics	General tool; no application to RTE
This work	RTE compliance audit	Sentinel-2, OSM, KSDB, WorldPop	RF + network Voronoi catchments	—

TABLE II
DATASETS USED

Dataset	Content used	Size
Sentinel-2 L2A [5]	Surface reflectance B4, B8, B11; Jan–Feb 2024 median composite, cloud < 5%	10 m raster
OSM walk network [7]	Pedestrian-accessible ways, private access excluded	186,595 nodes; 499,950 directed edges
OSM schools	amenity=school features with names	1,464 features
KSDB slums	Slum polygons with zone and notification status	509 polygons
WorldPop 2020 [8]	100 m population, summed per slum polygon	226,753 residents

F. Research Gap

Table I summarises the most relevant studies. Existing slum-mapping work stops at detection, and existing accessibility work assumes a complete population frame. To our knowledge, no published study for an Indian city combines open satellite classification, the official slum register, gridded population and full street-network routing to audit the RTE walking-distance mandate.

III. STUDY AREA AND DATA

A. Study Area

The study area is Bengaluru, Karnataka. Network routing and school features cover the OSM administrative boundary of the city. Satellite classification covers a 15 km radius around the city centre (12.9716°N, 77.5946°E). Fig. 1 shows the distribution of KSDB slum polygons, the classified sample locations, and the curated government schools.

B. Datasets

Table II lists the datasets. All are open or publicly released.

C. School Data and the Government-School Definition

OSM returned 1,464 school features (886 building or campus polygons and 578 points), of which 202 are unnamed. OSM ownership tags are sparse in Bengaluru, so government schools must be identified largely from names. The

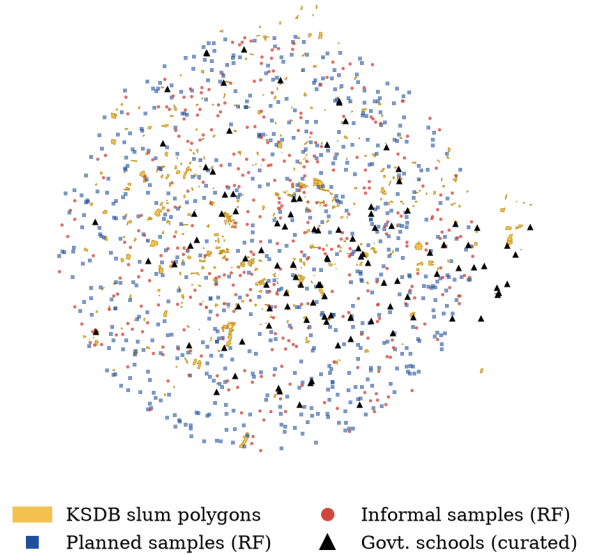


Fig. 1. Study area. KSDB slum polygons (yellow), Random Forest sample locations classified as planned (blue squares) and informal (red circles) within the 15 km classification radius, and the 111 curated government schools (black triangles).

initial pipeline used a *broad* filter: `operator:type` matching “public” or “government”, or a name containing “Govt”, “Government”, “BBMP” or “Public”. It selected 271 schools. Inspection showed that 150 of these carry no government keyword, and 125 matched only on “Public”. In Indian usage this word is common in private school names, and several of the schools in the broad set are visibly private institutions.

We therefore define a *curated* set of government schools. A school is included only if a government keyword (government, govt, BBMP, GHPS, GLPS, GHS, GULPS, KGBV, corporation or municipal) appears in the leading part of its name, i.e., not merely in an address suffix such as “, BBMP Agara”. Colleges and pre-university institutions are excluded. The curated set contains 111 schools, 49 of which are explicitly named as primary schools. The curated set is used for all primary results; the broad set is reported for comparison with

TABLE III
GOVERNMENT-SCHOOL IDENTIFICATION

Stage	Count
All OSM <code>amenity=school</code> features	1,464
Broad filter (initial pipeline)	271
of which matched only on "Public"	125
of which carry any government keyword	121
keyword only in address suffix (removed)	3
colleges / pre-university (removed)	7
Curated government schools	111
explicitly primary-named	49
Non-government schools (all others)	1,353

Mapped Schools in Bengaluru

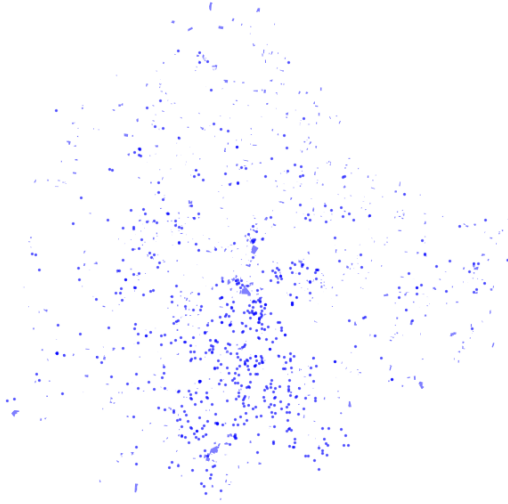


Fig. 2. All 1,464 OSM school features in Bengaluru (points and campus polygons).

the original pipeline (Table III).

IV. PROPOSED METHODOLOGY

Fig. 5 shows the four-phase pipeline. The phases are described below.

A. Phase 1: Data Acquisition and Pre-processing (Map)

Three ingestion streams initialise the pipeline. (i) A Sentinel-2 surface-reflectance median composite is built in GEE from scenes acquired between 1 January and 1 March 2024 (the dry season) with less than 5% cloudy pixels, clipped to the 15 km study radius. The median operator suppresses transient clouds and shadows. (ii) The pedestrian network is downloaded with OSMnx using `network_type="walk"` and a custom filter that excludes area features and ways tagged `access=private`. The graph has 186,595 nodes and 499,950 directed edges, with a median edge length of 40.4 m (mean 61.0 m). (iii) School features are fetched with `features_from_place` for `amenity=school`. Polygon features are reduced to centroids computed in the projected UTM zone 43N coordinate system (EPSG:32643) and then converted back to WGS-84.

Ground Truth: Schools vs. Informal Settlements in Bengaluru

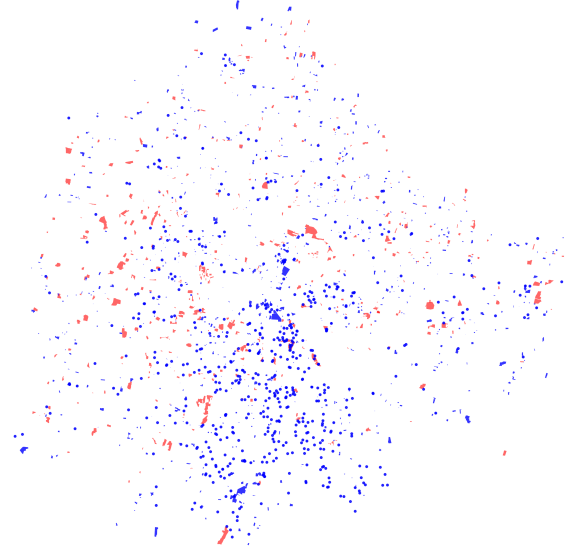


Fig. 3. Ground truth overlay: all OSM schools (blue) and KSDB informal-settlement polygons (red).

Corrected Ground Truth: Govt Schools vs. Informal Settlements

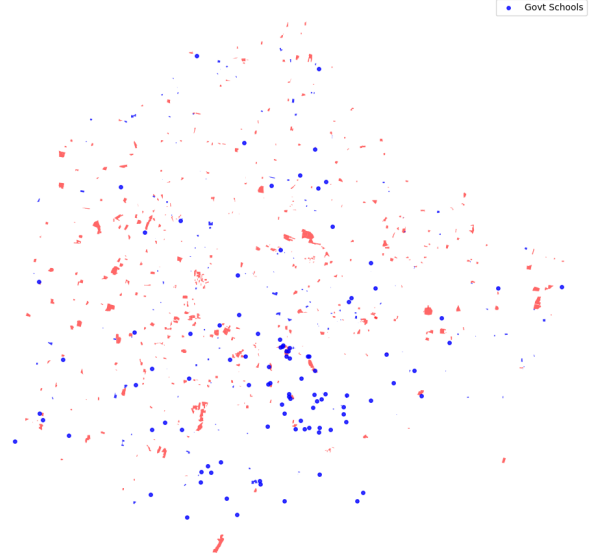


Fig. 4. Government schools selected by the broad filter of the initial pipeline (blue points) and KSDB informal-settlement polygons (red). The curated set used for the primary results is a subset of these schools.

B. Phase 2: Satellite Classification of Built-up Morphology (Detect)

Two spectral indices are computed from the composite:

$$\text{NDVI} = \frac{\rho_{\text{NIR}} - \rho_{\text{Red}}}{\rho_{\text{NIR}} + \rho_{\text{Red}}} = \frac{B8 - B4}{B8 + B4}, \quad (1)$$

$$\text{NDBI} = \frac{\rho_{\text{SWIR}} - \rho_{\text{NIR}}}{\rho_{\text{SWIR}} + \rho_{\text{NIR}}} = \frac{B11 - B8}{B11 + B8}. \quad (2)$$

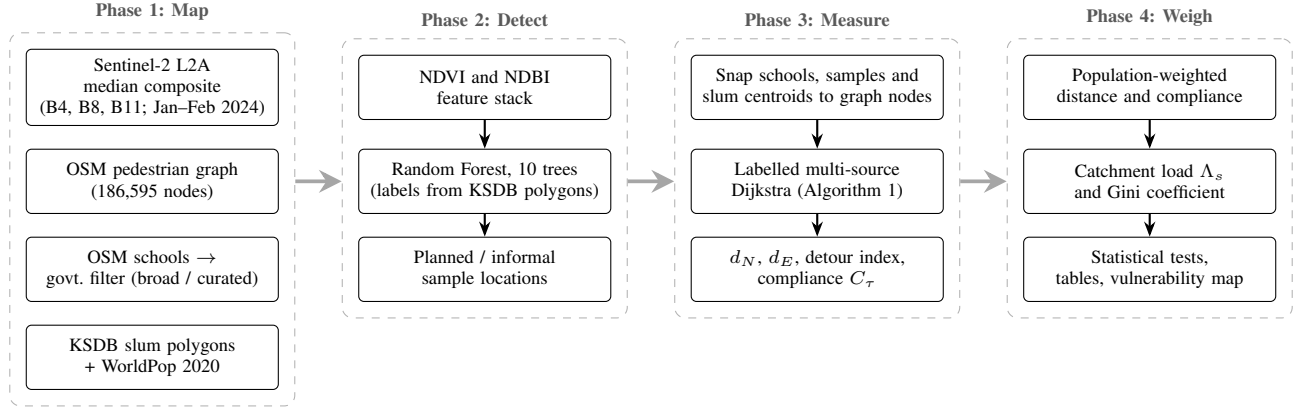


Fig. 5. Proposed pipeline. Phase 1 ingests open data; Phase 2 classifies built-up morphology from Sentinel-2 (training labels drawn from KSDB polygons); Phase 3 computes network distances with a labelled multi-source Dijkstra search; Phase 4 weights results by population and derives catchment loads.

TABLE IV
CLASSIFICATION CONFIGURATION

Parameter	Value
Imagery	Sentinel-2 SR (harmonised), median composite
Acquisition window	1 Jan – 1 Mar 2024
Cloud filter	CLOUDY_PIXEL_PERCENTAGE < 5
Study extent	15 km radius around city centre
Features	B4, B8, B11, NDVI, NDBI
Training pixels	2,000 informal + 2,000 planned
Classifier	Random Forest, 10 trees
Sampling scale	10 m
Output samples	364 informal, 636 planned

NDVI [17] suppresses vegetated pixels such as parks and tree-lined avenues; NDBI [18] emphasises impervious surfaces. The feature vector for each pixel is $\mathbf{x} = [B4, B8, B11, NDVI, NDBI]$.

Training data are drawn at 10 m scale. Class 1 (informal) consists of 2,000 pixels sampled inside KSDB slum polygons. Class 0 (planned) consists of 2,000 built-up pixels (NDBI > 0) sampled outside all slum polygons. A Random Forest [15] with 10 trees is trained in GEE and applied to the full study area. Sample locations (“virtual students”) are then drawn at random from each predicted class; the exported sample files contain 364 informal and 636 planned locations. Table IV lists the configuration.

C. Phase 3: Network Accessibility (Measure)

Let $G = (V, E, w)$ be the directed pedestrian graph with edge weights $w(e)$ equal to segment length in metres. Every origin p (a sample location or slum centroid) and every school $s \in S$ is snapped to its nearest graph node, $v(p)$ and $v(s)$. The network distance from p to the nearest government school is

$$d_N(p) = \min_{s \in S} \text{dist}_G(v(p), v(s)), \quad (3)$$

and the corresponding straight-line distance, computed in UTM coordinates, is

$$d_E(p) = \min_{s \in S} \|\mathbf{u}(p) - \mathbf{u}(s)\|_2. \quad (4)$$

Algorithm 1: Labelled multi-source Dijkstra

Input: graph $G = (V, E, w)$; school nodes $s_1, \dots, s_k$
Output: distance $D[v]$ and nearest-school label $L[v]$ for all reachable v

- 1) $D[v] \leftarrow \infty$ for all $v \in V$; priority queue $Q \leftarrow \emptyset$
- 2) **for** $i = 1$ **to** k : **if** $D[s_i] = \infty$: $D[s_i] \leftarrow 0$; $L[s_i] \leftarrow i$; push $(0, s_i, i)$ to Q
- 3) **while** $Q \neq \emptyset$:
- 4) pop (d, u, i) with minimum d
- 5) **if** $d > D[u]$: **continue** (stale entry)
- 6) **for each** $(u, v) \in E$:
- 7) **if** $d + w(u, v) < D[v]$:
- 8) $D[v] \leftarrow d + w(u, v)$; $L[v] \leftarrow i$; push $(D[v], v, i)$
- 9) **return** D, L

Computing d_N by running one shortest-path search per origin would be wasteful. Instead, we run a single multi-source Dijkstra search [25] seeded simultaneously from all school nodes. Propagating the index of the seeding school as a *label* alongside the tentative distance yields, on termination, both the distance to the nearest school and its identity at every node (Algorithm 1). This is the standard construction of the graph (network) Voronoi diagram [23], [24]; each Voronoi cell is the walking catchment of one school. The search runs in $O((|V| + |E|) \log |V|)$ time; on the Bengaluru graph a full pass takes a few seconds on a laptop.

The *detour index* of an origin is $\eta(p) = d_N(p)/d_E(p)$ (with d_E floored at 50 m to avoid division by near-zero distances). The RTE compliance rate of a set of origins P at threshold τ is

$$C_\tau(P) = \frac{1}{|P|} \sum_{p \in P} \mathbf{1}[d_N(p) \leq \tau], \quad (5)$$

with $\tau = 1,000 m$ for primary schooling. The crow-flies compliance rate C_τ^E is defined analogously with d_E . The *false-compliance* share is the fraction of crow-flies-compliant origins that are not network-compliant.

D. Phase 4: Population Weighting and Catchment Load (Weigh)

For each KSDB slum k with WorldPop population π_k , the population-weighted network distance and population compli-

ance are

$$\bar{d}_\pi = \frac{\sum_k \pi_k d_N(c_k)}{\sum_k \pi_k}, \quad C_\tau^\pi = \frac{\sum_k \pi_k \mathbf{1}[d_N(c_k) \leq \tau]}{\sum_k \pi_k}, \quad (6)$$

where c_k is the slum centroid. Using the labels from Algorithm 1, the *catchment load* of school s is the slum population for which s is the nearest government school by network:

$$\Lambda_s = \sum_k \pi_k \mathbf{1}[L[v(c_k)] = s]. \quad (7)$$

Inequality of loads across schools is summarised with the Gini coefficient

$$\mathcal{G} = \frac{\sum_i \sum_j |\Lambda_i - \Lambda_j|}{2|S|^2 \bar{\Lambda}}. \quad (8)$$

Because school enrolment capacity is not available for Bengaluru in open form, Λ_s measures the *potential* demand that falls on each school, not its utilisation.

E. Statistical Analysis

Network distances are right-skewed, so formal and informal samples are compared with the two-sided Mann–Whitney U test [30]. Effect size is reported as Cliff’s δ [31], interpreted with the thresholds of Romano *et al.* [32] ($|\delta| < 0.147$ negligible, < 0.33 small, < 0.474 medium). The full distributions are compared with the two-sample Kolmogorov–Smirnov test. Compliance proportions are compared with a χ^2 test. A 95% confidence interval for the difference in means is obtained by the percentile bootstrap with 5,000 resamples [33].

F. Implementation

Classification runs in GEE through its Python API and geemap. Routing and analysis use Python 3 with OSMnx 2.1, NetworkX, GeoPandas, SciPy, and Matplotlib. Every number and figure in Section V is regenerated by a single script, `scripts/paper_analysis.py`, which runs in about 20 s after the graph is cached.

V. RESULTS

A. Walking Distance from Classified Locations

Table V summarises the network and straight-line distances from the 1,000 sample locations to the nearest curated government school. Distances are long in both zone types. The median planned location is 2.07 km from a government school by network and the median informal location is 1.92 km. Only 16.5% of planned and 18.7% of informal locations are within the 1 km RTE neighbourhood, and roughly half of each group is more than 2 km away. Fig. 6 compares the means under both school definitions, and Fig. 7 shows the full cumulative distributions.

B. Formal versus Informal: How Large Is the Difference?

Table VI reports the statistical comparison. With curated schools, planned locations are on average 220 m farther than informal locations (95% bootstrap CI: 41–382 m). However, the rank-based Mann–Whitney test is not significant at the 5% level ($p = 0.094$), the Kolmogorov–Smirnov test finds no

TABLE V
DISTANCE TO NEAREST GOVERNMENT SCHOOL (CURATED SET, $n = 111$ SCHOOLS)

Statistic	Network d_N (m)		Straight line d_E (m)	
	Planned	Informal	Planned	Informal
n	636	364	636	364
Mean	2,318	2,098	1,662	1,503
Std. dev.	1,557	1,164	1,041	913
Median	2,070	1,919	1,448	1,316
Q1 – Q3	1,239–3,024	1,244–2,884	848–2,245	813–1,982
90th pct.	4,207	3,677	3,231	2,764
≤ 1 km (%)	16.5	18.7	31.8	35.2
> 2 km (%)	52.7	48.1	31.8	24.5
Median detour η	1.35	1.35	—	—

Median snapping distance: 37.6 m (planned), 47.0 m (informal), 28.8 m (schools).

TABLE VI
PLANNED VS. INFORMAL COMPARISON UNDER TWO GOVERNMENT-SCHOOL DEFINITIONS

Quantity	Curated (111)	Broad (271)
Mean d_N , planned (m)	2,318	1,548
Mean d_N , informal (m)	2,098	1,351
Mean difference (m)	220	197
95% bootstrap CI (m)	[41, 382]	[81, 314]
Mann–Whitney U	123,120	128,255
p -value	0.094	0.004
Cliff’s δ	0.06 (negl.)	0.11 (negl.)
KS statistic D (p)	0.072 (0.18)	0.091 (0.04)
χ^2 on 1 km compliance (p)	0.62 (0.43)	5.47 (0.02)
Compliance ≤ 1 km, planned (%)	16.5	32.5
Compliance ≤ 1 km, informal (%)	18.7	40.1

difference in distributions ($p = 0.18$), the compliance rates do not differ (χ^2 $p = 0.43$), and Cliff’s $\delta = 0.06$ is negligible. With the broad filter, the same direction appears more strongly ($p = 0.004$, $\delta = 0.11$), but that filter counts private schools as government schools.

The direction of the gap is consistent: planned layouts are, if anything, farther from government schools than informal areas. The magnitude, however, is small compared with the distance both groups must walk. An earlier run of the pipeline on a different OSM snapshot with a 257-school broad filter produced means of 1.60 km (planned) and 1.43 km (informal), which is consistent with the broad-filter column of Table VI.

C. Sensitivity to the Distance Threshold

Fig. 8 and Table VII show compliance as a function of the threshold τ . Compliance rises steeply between 1 and 3 km. At 3 km, the RTE neighbourhood for upper-primary schooling (Classes VI–VIII), 74.4% of planned and 79.9% of informal locations are covered. The shortfall is therefore most severe for the youngest children, for whom the strictest (1 km) standard applies.

D. The Crow-Flies Fallacy

Fig. 9 plots network against straight-line distance for every sample location, and Table VIII summarises the effect on compliance. Network and Euclidean distances are strongly

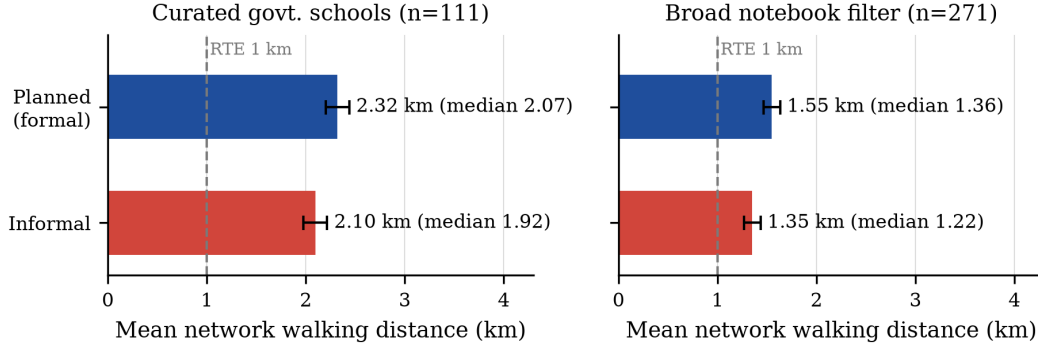


Fig. 6. Mean network walking distance to the nearest government school, with 95% bootstrap confidence intervals. Left: curated government schools (primary analysis). Right: broad filter of the initial pipeline, which includes many private “Public” schools. The dashed line marks the 1 km RTE neighbourhood.

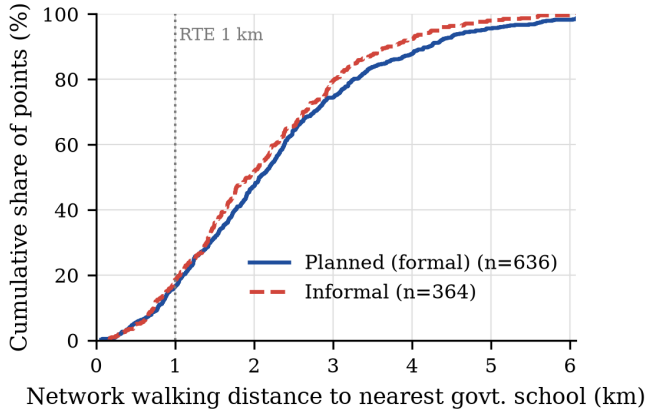


Fig. 7. Empirical cumulative distribution of network walking distance to the nearest curated government school. The two curves are close throughout; the informal curve lies slightly to the left.

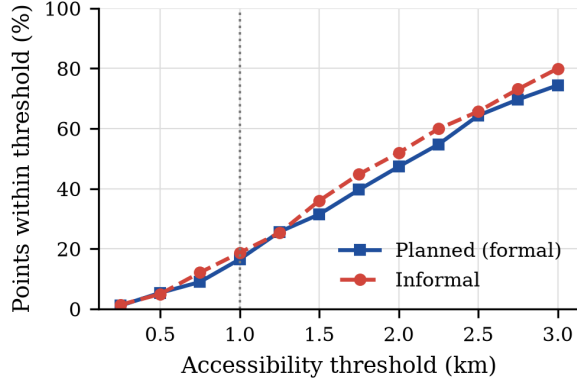


Fig. 8. Share of sample locations within a given network distance of a curated government school. The dotted line marks the 1 km primary-school neighbourhood.

rank-correlated (Spearman $\rho = 0.93$), but the median detour index is 1.35: a typical walk is roughly one-third longer than the straight line. At the 1 km threshold this matters a great deal. A straight-line audit would classify 33.0% of locations

TABLE VII
COMPLIANCE (%) BY NETWORK-DISTANCE THRESHOLD

Threshold	0.5 km	1 km	1.5 km	2 km	3 km
Planned	5.3	16.5	31.4	47.3	74.4
Informal	4.9	18.7	36.0	51.9	79.9

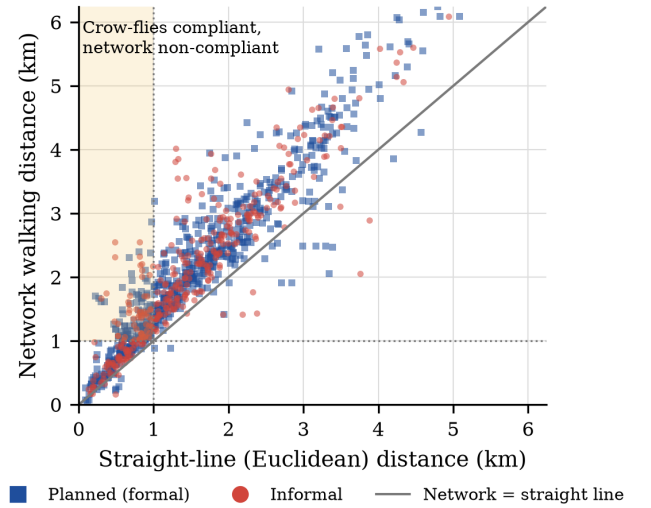


Fig. 9. Network vs. straight-line distance to the nearest curated government school. Points in the shaded region lie within 1 km in a straight line but more than 1 km by the street network.

as compliant, whereas the network audit finds 17.3%. Of the locations that a crow-flies buffer would certify as compliant, 48.2% are actually more than 1 km away on foot. With the broad school set the pattern is the same: 58.9% vs. 35.3%, a false-compliance share of 41.4%.

E. Population-Weighted Access for KSDB Slums

The classified sample locations depend on the satellite classifier (see Section VII). The KSDB slum polygons offer an independent, register-based view. Table IX reports network access for all 509 polygons, which together hold an esti-

TABLE VIII
STRAIGHT-LINE VS. NETWORK COMPLIANCE AT 1 KM

Quantity	Curated	Broad
Crow-flies compliant, C_{1km}^E (%)	33.0	58.9
Network compliant, C_{1km} (%)	17.3	35.3
False-compliant, share of all locations (%)	15.9	24.4
False-compliant, share of crow-flies compliant (%)	48.2	41.4
Spearman $\rho(d_N, d_E)$	0.93	0.87

TABLE IX
NETWORK ACCESS FOR KSDB SLUMS (WORLDPop-WEIGHTED)

Quantity	Curated	Broad
Slums routed	509	509
Residents (WorldPop 2020)	226,753	226,753
Median slum distance (m)	1,723	1,103
Population-weighted distance (m)	2,023	1,331
Slums within 1 km (%)	27.7	45.8
Residents within 1 km (%)	20.1	39.2
Slums beyond 2 km (%)	39.1	11.6
Spearman ρ (population, distance)	0.06 ($p=0.15$)	0.05 ($p=0.28$)

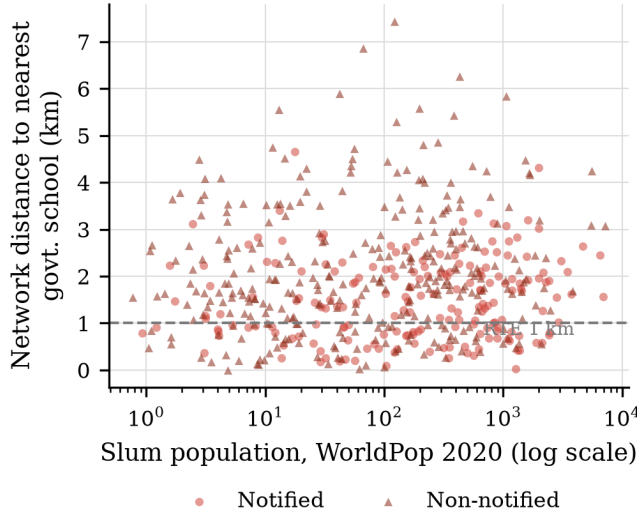


Fig. 10. Network distance to the nearest curated government school vs. WorldPop population for each KSDB slum, by notification status. Settlement size does not predict proximity.

mated 226,753 residents (median slum: 1.25 ha, 127 residents, 187 residents/ha). With curated schools, the median slum centroid is 1.72 km from a government school by network. The population-weighted distance is 2.02 km, and only 20.1% of slum residents live within the 1 km RTE neighbourhood. 39.1% of slums are more than 2 km away.

Fig. 10 plots each slum’s distance against its population. There is no relationship between settlement size and distance ($\rho = 0.06$, $p = 0.15$), nor between density and distance ($\rho = -0.02$, $p = 0.59$). Large slums are no better served than small ones.

1) *Notified versus non-notified slums:* Table X separates slums by notification status. Non-notified slums, i.e., set-

TABLE X
ACCESS BY SLUM NOTIFICATION STATUS (CURATED SCHOOLS)

Quantity	Notified	Non-notified
Slums	192	317
Residents	122,705	104,049
Median distance (m)	1,441	1,862
Population-weighted distance (m)	1,707	2,396
Slums within 1 km (%)	35.9	22.7
Residents within 1 km (%)	24.0	15.5

TABLE XI
SLUM ACCESS BY ADMINISTRATIVE ZONE (CURATED SCHOOLS)

Zone	Slums	Residents	Pop.-wtd. dist. (m)	Res. ≤ 1 km (%)
South	93	48,486	1,167	43.8
Mahadevapura	90	1,366	1,507	35.8
West	90	67,150	1,889	17.9
East	103	54,730	2,079	17.6
RR Nagar	55	24,690	2,443	4.4
Yelahanka	46	11,101	3,170	8.6
Bommanahalli	14	325	3,174	10.1
Dasarahalli	18	18,906	3,330	0.7

tlements on the register but without legal notification, are significantly farther from government schools than notified ones. The population-weighted distance is 2.40 vs. 1.71 km, and only 15.5% of non-notified residents are within 1 km, compared with 24.0% of notified residents (Mann–Whitney $p = 1.5 \times 10^{-6}$, Cliff’s $\delta = 0.25$, small). The difference persists with the broad school set ($p = 0.002$, $\delta = 0.16$).

2) *Administrative zones:* Access varies sharply between the city’s administrative zones (Table XI, Figs. 11 and 12). South zone is best served: 43.8% of its slum residents are within 1 km, at a population-weighted distance of 1.17 km. Dasarahalli, Yelahanka and Bommanahalli are the worst: population-weighted distances exceed 3 km, and in Dasarahalli fewer than 1% of slum residents are within 1 km of a curated government school. The peripheral zones, where much post-2011 growth has occurred, are the least served. The spread within zones is also wide (Fig. 12): even in the best-served South zone, a quarter of slums are more than 1.7 km from a government school, while in Dasarahalli the median slum is 3.4 km away.

F. Catchment Load: Where Slum Demand Falls

Algorithm 1 assigns every slum to its nearest government school. Of the 111 curated schools, 81 are the nearest school for at least one slum, and 30 serve none. The distribution of potential demand is extremely unequal (Fig. 13, Table XII). The median school with a slum catchment receives 179 slum residents, the mean is 2,799, and the most-loaded school receives 28,193. The ten most-loaded schools are the nearest government option for 71.5% of all slum residents, and the Gini coefficient of loads is 0.85. With the broad filter the Gini is similar (0.87), but the most-loaded “government” schools are private institutions with “Public” in their names. This

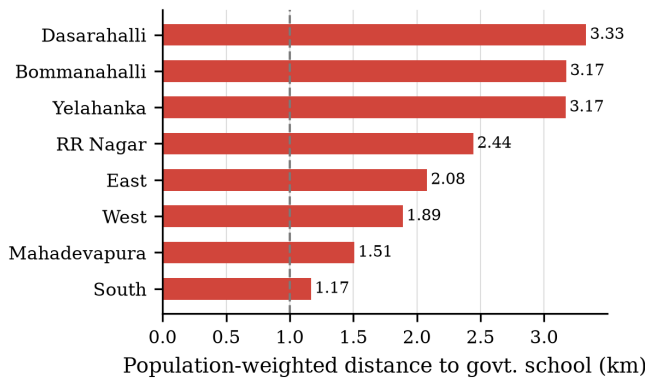


Fig. 11. Population-weighted network distance from KSDB slums to the nearest curated government school, by administrative zone.

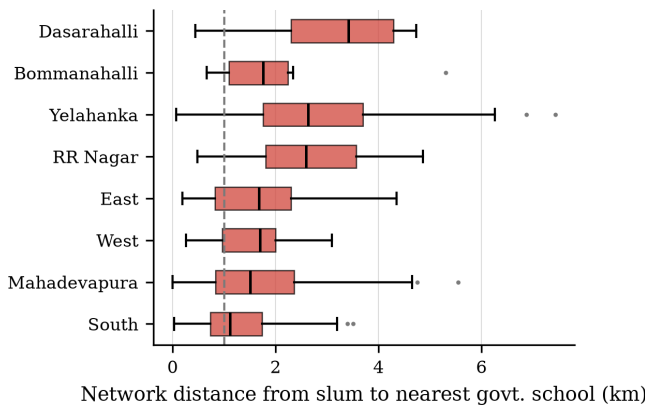


Fig. 12. Distribution of network distance from each KSDB slum to the nearest curated government school, by administrative zone (boxes: interquartile range; line: median; dashed line: 1 km RTE neighbourhood).

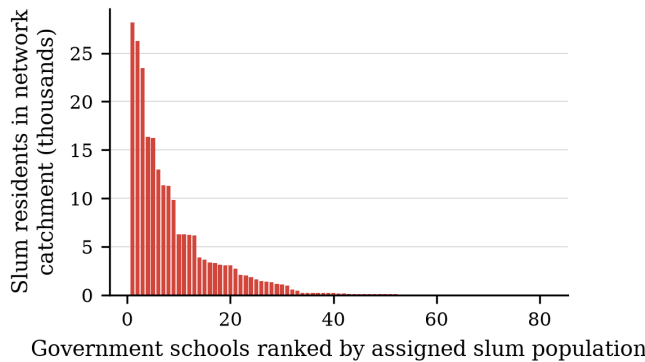


Fig. 13. Slum residents for whom each curated government school is the nearest by network, ranked. A small number of schools absorb most potential slum demand.

illustrates how misclassification can shift apparent demand onto schools that do not provide free education.

TABLE XII
MOST-LOADED GOVERNMENT SCHOOLS (CURATED SET)

School (OSM name)	Slums	Residents
Haji Sir Ismail Sait Govt. Urdu & English HPS	25	28,193
Nammura Government Model Primary School	22	26,242
Government School (generic OSM name)	27	23,453
Gangamma Thimmaiah Govt. School	14	16,339
Jindal Government School	10	16,223
Top-10 share of slum residents	71.5%	
Gini coefficient of loads	0.85	

TABLE XIII
NETWORK DISTANCE TO NEAREST GOVERNMENT VS.
NON-GOVERNMENT SCHOOL

	Govt. (curated)		Non-govt.	
	Plan.	Inf.	Plan.	Inf.
Median (m)	2,070	1,919	691	598
Mean (m)	2,318	2,098	858	757
≤ 1 km (%)	16.5	18.7	71.2	76.4
> 2 km (%)	52.7	48.1	5.3	4.7

G. Proximity to Non-Government Schools

The 1,353 OSM school features outside the curated government set are much closer to every location (Table XIII). The median planned location is 0.69 km from a non-government school and the median informal location 0.60 km; 71.2% and 76.4% respectively are within 1 km. The open data therefore describe a city in which a school of *some* kind is usually within walking distance, but a free government school usually is not.

H. Vulnerability and Catchment Maps

Fig. 14 maps compliance across the city using the curated government schools: (a) the classified sample locations and (b) the KSDB slums, each coloured by network walking distance. Non-compliant locations dominate the periphery, and the large slums in the West and RR Nagar zones form the most extensive cluster beyond 2 km.

Fig. 15 visualises the network Voronoi partition produced by Algorithm 1. Every street node is coloured by the government school it is nearest to, and nodes more than 1 km from any government school are faded. The RTE-compliant area appears as isolated islands around each school rather than a continuous surface, and large parts of the western and northern city—where the biggest slums are concentrated—lie outside every island.

I. Interactive Web Map

The pipeline also produces an interactive web map (Leaflet via folium) in which every sample location is coloured by its walking distance: green (≤ 1 km, compliant), orange (1–2 km) and red (> 2 km, critical). Each marker's pop-up reports its zone type and distance, and government schools are shown as small blue points. Figs. 16 and 17 show two views of the interface. These screenshots come from the initial pipeline and

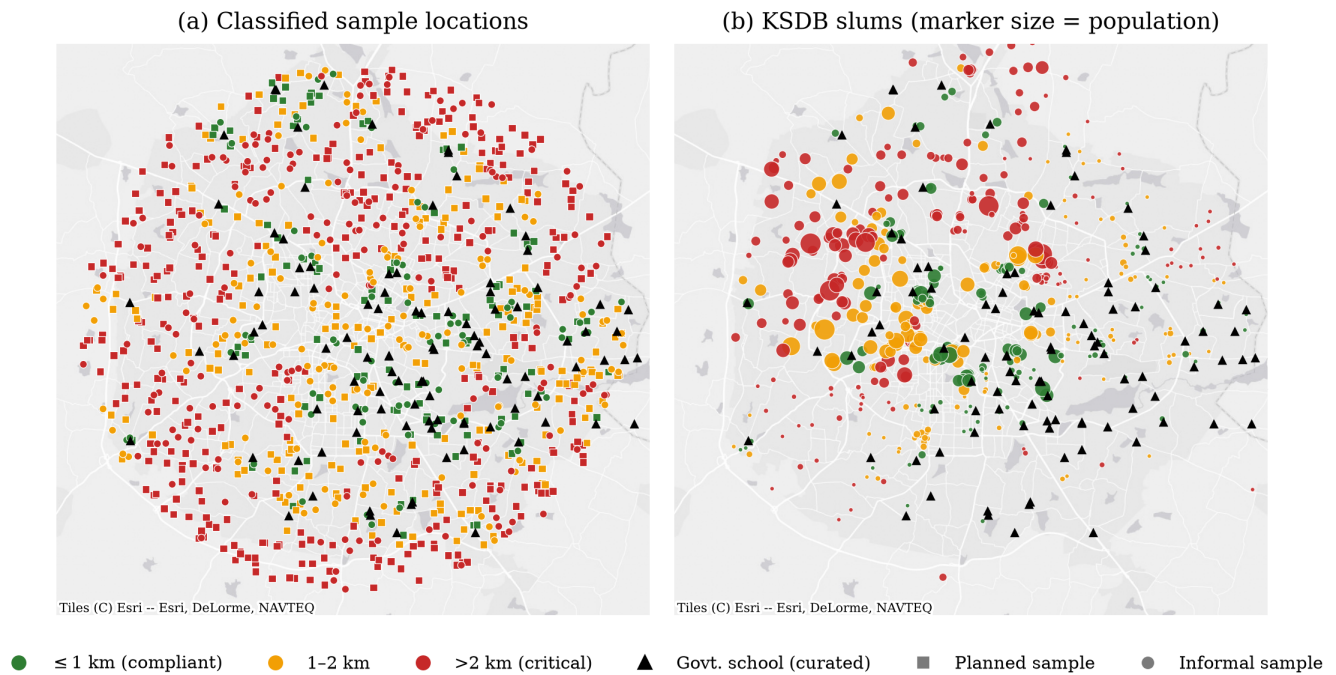


Fig. 14. Vulnerability map of Bengaluru (curated government schools). Colour gives network walking distance to the nearest government school: green ≤ 1 km (compliant), orange 1–2 km, red > 2 km (critical). (a) Random Forest sample locations (squares: planned; circles: informal). (b) KSDB slum centroids, sized by WorldPop population. Black triangles are curated government schools.

therefore use the broad school filter; an equivalent map built on the curated set is exported alongside the analysis.

VI. DISCUSSION

A. A City-Wide Compliance Gap

The most robust finding does not depend on the formal–informal distinction: across Bengaluru, most children cannot reach a government school within the 1 km walk the RTE rules promise. This holds for both classified zone types, for registered slums, and under both school definitions. Only the size of the gap changes, from roughly 60–65% non-compliance with the broad set to over 80% with the curated set. For the youngest children, the effective choice is between a long walk to a government school and a nearby private school.

B. Planned Layouts and Privatised Substitution

Planned layouts are, on average, slightly farther from government schools than informal areas, and they are close to dense networks of non-government schools. This pattern is consistent with *privatised substitution*: where households can pay, demand shifts to private schools [28], and there is little pressure to site government schools in planned layouts. Policy changes in Karnataka point the same way. A January 2019 amendment to Rule 4 of the Karnataka RTE Rules exempted private unaided schools from the 25% RTE quota in neighbourhoods that have a government or aided school, and the Karnataka High Court upheld the amendment in May 2019 [35]. RTE-quota admissions have since been reported to have fallen from over one lakh in 2018 to 3,412 [36]. Under

this rule, the presence of a government school, however far away on foot, determines whether a poor child can claim a free seat in a nearby private school, which makes accurate walking-distance audits all the more consequential. Our data cannot establish causality, and the formal–informal difference we measure is small. The hypothesis that planned layouts are *strongly* disadvantaged is not supported once private schools are removed from the government set.

C. Invisibility as Exclusion

The clearest equity signal in the data concerns administrative visibility. Non-notified slums are significantly farther from government schools than notified slums, and peripheral zones are the least served. These are the settlements least likely to appear in the planning frames used for conventional RTE audits. A network audit built from open data can expose this gap directly, which is the central practical argument for the pipeline.

D. From Distance to Load: Re-examining the Density Penalty

An earlier version of this work proposed a “density penalty” in which proximity in informal settlements is offset by overcrowding. The real data refine that idea. Settlement population and density do not predict distance. However, potential demand from slums is concentrated on a small number of government schools: ten schools are the nearest option for over 70% of slum residents. Without open enrolment and capacity data, we cannot say whether these schools are over capacity. The load ranking nevertheless gives planners a short, concrete

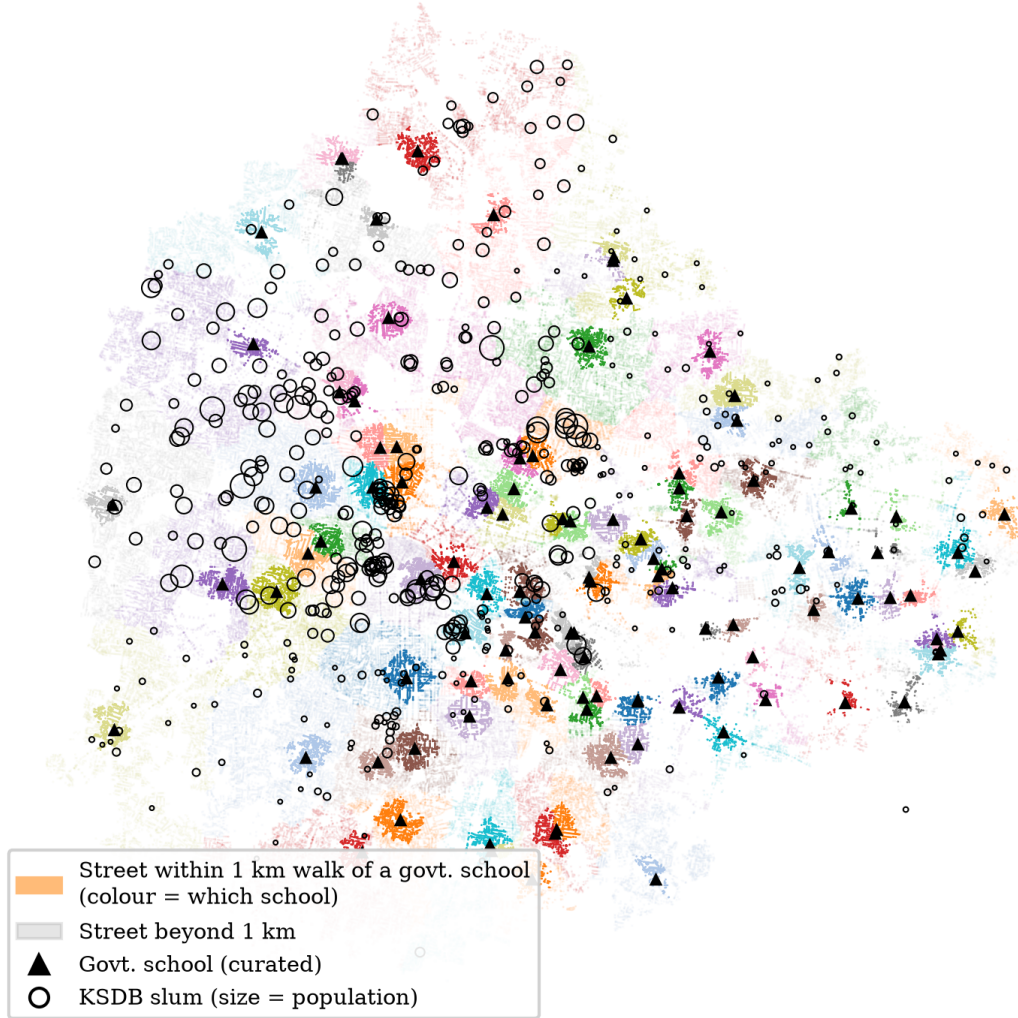


Fig. 15. Network Voronoi catchments of the 111 curated government schools on the Bengaluru pedestrian network. Each street node is coloured by its nearest government school (colours repeat and only distinguish neighbouring catchments); nodes beyond a 1 km walk are faded. Open circles are KSDB slums sized by population; black triangles are schools.

list of schools whose capacity should be checked first, and it can feed directly into capacity-aware measures such as 2SFCA [20] once capacities are known.

E. Implications for Auditing Practice

Three recommendations follow. First, compliance audits should use network distance: straight-line buffers misclassify about half of the apparently compliant locations. Second, audits should include the full slum register, not only notified slums. Third, the identity of a “government school” in open data needs curation; a naive keyword filter roughly halves the apparent gap.

VII. THREATS TO VALIDITY AND LIMITATIONS

A. Agreement between the Classifier and KSDB Ground Truth

The Random Forest was trained on pixels inside KSDB polygons, yet its predicted classes agree poorly with those polygons at the sample locations (Table XIV). Only 1.9% of locations predicted informal fall inside a KSDB polygon (vs. 0.8% of planned locations), and within 250 m or 500 m buffers the two classes are indistinguishable. Several factors contribute. KSDB slums are small (median 1.25 ha, i.e., about 125 Sentinel-2 pixels), so at 10 m resolution many slum pixels are spectrally mixed with their surroundings, as Wurm *et al.* [14] also found. The KSDB register is also incomplete, and the model uses only five spectral features with no texture. Formal accuracy assessment on held-out labelled data was not performed. Consequently, the “informal” class should be read

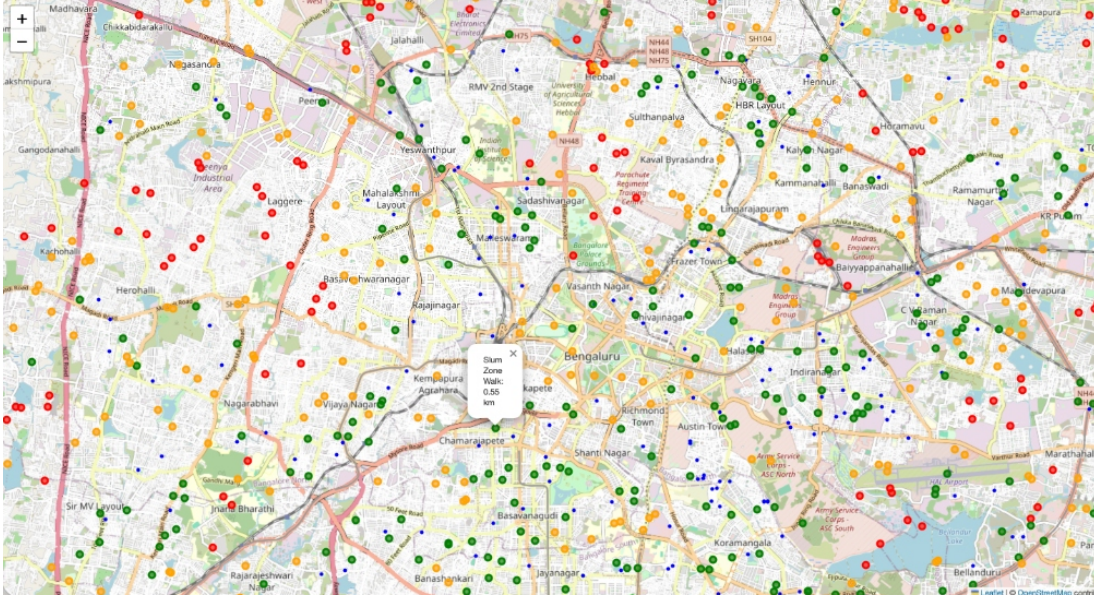


Fig. 16. Interactive vulnerability map of central Bengaluru (broad school filter), with an informal-zone location 0.55 km from the nearest government school selected. Markers show sample locations coloured by network walking distance: green ≤ 1 km, orange 1–2 km, red > 2 km; small blue points are schools.

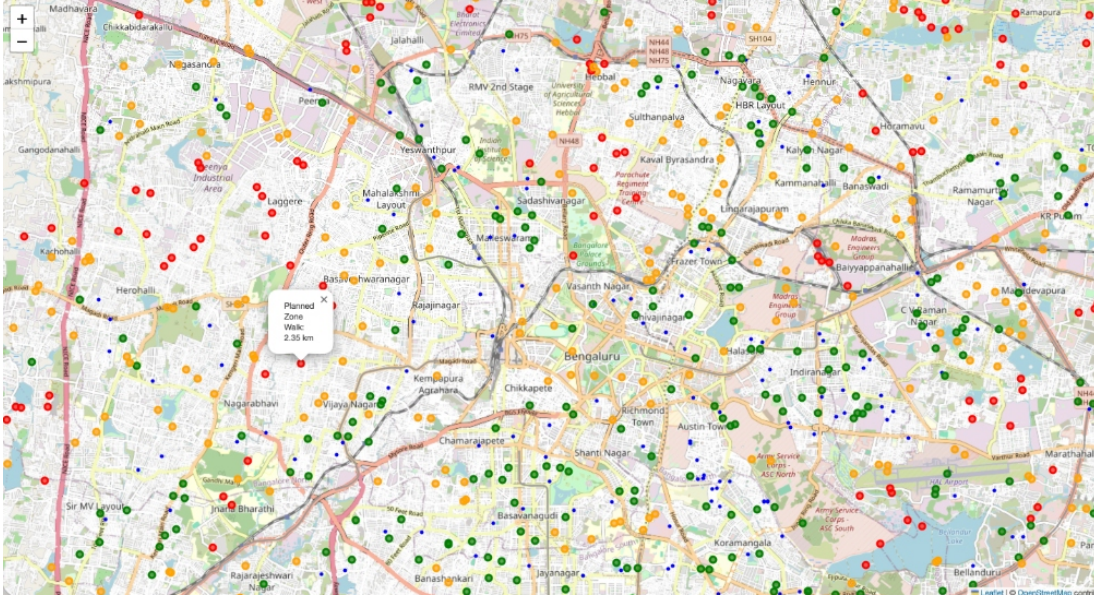


Fig. 17. The same vulnerability map with a planned-zone location 2.35 km from the nearest government school selected.

as *informal-like spectral morphology*, not as verified slum. This is why we report the KSDB-based results (Section V-E) alongside the classified-sample results, and why our equity conclusions rest primarily on the former.

B. School Data

OSM school coverage and attributes are incomplete. Government schools that are unmapped or unnamed in OSM are missing from both sets, which biases distances upward. The curated filter trades recall for precision: it removes private “Public” schools but may also drop government schools whose names lack a keyword. The broad and curated results therefore

TABLE XIV
AGREEMENT OF CLASSIFIED SAMPLES WITH KSDB SLUM POLYGONS

Share of samples (%)	Pred. informal	Pred. planned
Inside a KSDB polygon	1.9	0.8
Within 100 m of a polygon	6.9	6.1
Within 250 m of a polygon	15.9	16.4
Within 500 m of a polygon	41.2	39.6
<i>n</i>	364	636

bracket the plausible range. Neither set distinguishes reliably

between primary and secondary levels; 49 of the 111 curated schools are explicitly primary. Because missing government schools can only lengthen measured distances, all compliance figures in this paper are best read as lower bounds on access. Official UDISE+ records would remove this uncertainty, but they could not be used to validate OSM coverage: the public UDISE+ dashboard and reports (e.g., Report 1003, schools by management and category) are disaggregated only to state level for 2025–26, reporting 74,578 schools in Karnataka of which 65.07% are government-managed, with no district- or city-level breakdown [34], and school coordinates are available only to authorised users of the UDISE+ GIS portal. The completeness of OSM’s government-school layer for Bengaluru therefore cannot be quantified from open data.

C. Routing Assumptions

Distances are measured node to node; the snapping leg (median 22–47 m) is excluded. Polygon schools are represented by centroids rather than gates, which can over- or under-state the walk to large campuses. The OSM walk network may omit informal footpaths through settlements, which would overstate distances for informal origins, and it does not model road-crossing difficulty or safety, which would understate effective distances.

D. Population Data

WorldPop 2020 is a modelled product that disaggregates census totals using covariates, and it is known to be uncertain in dense informal areas. Slum populations here are therefore estimates. The catchment load counts all residents, not school-age children, and does not reflect actual enrolment choices.

VIII. CONCLUSION AND FUTURE WORK

This paper presented an open, reproducible pipeline for auditing the RTE walking-distance mandate. It combines Sentinel-2 classification, OSM network routing with a labelled multi-source Dijkstra search, the KSDB slum register, and WorldPop population. Applied to Bengaluru, it shows that the compliance gap is city-wide: with curated government schools, only 17.3% of sampled locations and 20.1% of slum residents are within a 1 km walk. Planned layouts are marginally farther from government schools than informal areas, but the difference is negligible in effect size and sensitive to how government schools are identified. Straight-line audits overstate compliance, with nearly half of apparently compliant locations failing on the street network. Non-notified slums and peripheral zones are the least served, and potential slum demand concentrates on a handful of government schools.

Future work will address the limitations above:

- 1) **Authoritative school data.** Replace name-based filtering with UDISE+ school locations, management type and level (obtained through authorised access to the UDISE+ GIS portal), and add enrolment capacities to compute capacity-aware 2SFCA accessibility.
- 2) **Validated classification.** Add GLCM texture [13] and VHR imagery, create an independent labelled validation

set, and report a proper ablation with accuracy, precision, recall and F1-score.

- 3) **School-age population.** Use age-structured WorldPop layers to weight by children aged 6–14 rather than all residents.
- 4) **Public transit.** Integrate BMTC bus stops and routes to assess whether locations beyond walking range have viable transit access to schools.
- 5) **Siting optimisation and dashboard.** Deploy the pipeline as an interactive planning dashboard and add location–allocation optimisation to recommend sites for new government schools that maximise population-weighted 1 km coverage under budget constraints.

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